

Figure 1. Foods that are rich in antioxidants.

Credit: fcafotodigital, Getty Images

Antioxidants are substances that interact with highly reactive species in the body called radicals, preventing the radicals from causing cellular damage. These radicals play a role in normal biological function and certain diseases. Radicals are produced in the body through normal metabolic processes and can cause cellular damage that may lead to disease. Eating a diet rich in fruits and vegetables provides the body with a variety of antioxidant species to react with these excess radicals.

Nature of Science**Aspect: Evidence**

More is not always better! Diets rich in fruits and vegetables provide the human body with important antioxidants to combat cellular damage caused by radicals. Consuming high doses of antioxidants via supplements, however, has not shown to be beneficial in preventing disease and in some cases has increased risk. This [fact sheet](https://www.nccih.nih.gov/health/antioxidants-in-depth)  (<https://www.nccih.nih.gov/health/antioxidants-in-depth>) from the US National Institutes of Health reviews the scientific evidence of antioxidant supplements and why they don't seem to be effective in preventing disease.

Many lifestyle factors such as smoking, exposure to air pollution, ultraviolet radiation and pesticides, consumption of fried and processed foods can accelerate the production of radicals in the body. This is one reason why these lifestyle factors are linked to many diseases. It truly is amazing how the chemistry of the foods we eat can have such a profound impact on our overall health.

Creativity, activity, service**Strand: Activity****Learning outcomes:**

- Show commitment to, and perseverance in CAS experiences
- Identify own strengths and develop areas for growth

It is estimated that lifestyle diseases 

(<https://www.un.org/en/chronicle/article/lifestyle-diseases-economic-burden-health-services>), also known as chronic diseases, will cause a staggering 70% of global deaths by 2030. Such diseases are caused by a lack of physical activity and unhealthy eating habits. Keeping fit and active is important and can help you maintain a healthy mind and body.

Possible CAS experiences that address the activity strand include working out at the gym, swimming or taking part in yoga classes. Make sure that you take time to enjoy activities like these as part of your busy schedule as an IB student.

Prior learning

Before you study this subtopic make sure that you understand the following:

- The names and structures of alkanes and halogenoalkanes (see section S3.2.3—4  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>)).
- The nature of the covalent bond (see section S2.2.1a  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>)).
- How to deduce the Lewis formula of molecules and ions (see section S2.2.1a  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>)).